DEL MAR
EST. SEAFOODS 1988

NOTES FROM QINGDAO · 青岛会谈备忘

PLATE FREEZER PROJECT NOTES

平板冻结项目备忘

What we talked about and what we're after, so Wolfking can get started. English first, then Chinese.
我们谈到的内容和我们想要的东西，方便狼王着手。先英文，后中文。

DEL MAR SEAFOODS · 30 OCTOBER 2026 · FOR WOLFKING AND APCCO

ENGLISH

WHO'S DOING WHAT

- **Del Mar (Joe)** — the customer. Picks the block size, approves the layout and the final design.
- **Wolfking** — designs and builds four vertical plate freezers with hydraulics and controls, plus the ammonia refrigeration package to run them. Sends drawings and the paperwork the permits need, and a couple of technicians to help APCCO's crew install and start up.
- **APCCO** — our local refrigeration contractor in California. Checks the design against California rules, stamps and files the permits, installs everything (piping, insulation, electrical) and services it afterwards. They don't build freezers; they're here to get it installed and legal. They'll talk straight to Wolfking's engineers and will ask a lot of questions. That's just the permit process.
- Let's keep everything on the one email thread with all three of us copied.

> **Sean, about the Poland story** (papers demanded at the port): before anything is built, APCCO will send Wolfking a list of the certificates and documents the permit will need, and Wolfking tells us which ones it can provide. No surprises at the port.

WHAT WE'RE MAKING

- **Whole California squid** in 10 kg blocks. A test for the China market: if Chinese buyers like plate-frozen squid, we convert more of our production. Squid season on Monterey Bay starts in May, so we want to be running by May 2027.
- **Ground salmon racks** (the frames left after filleting, ground to a paste) for a pet-food customer, year-round. That contract is what pays for the machines.

45 t	2 × 10 h	4	140
a day (50 short tons)	shifts + 4 h cleaning	vertical plate freezers	blocks of 10 kg per batch, per freezer (the wish)

Our math: 90 min freeze + 30 min unload/reload = a batch every 2 hours, 4 per shift, 8 a day. 45 t ÷ 8 ÷ 4 freezers ≈ 1.4 t, or 140 blocks, per freezer per batch. If the design lands a bit lower, that's okay.

Why we're doing this: today we blast-freeze squid in cartons on racks, which is a lot of hand labor at $30 an hour. These four freezers are a test. If it works, we add more of the same design, and later we automate loading, unloading, bagging and boxing.

THE BLOCK

- 10 kg, about 12 litres (12,000,000 mm^3). Squid is roughly 20% lighter than water, so 12 L ≈ 10 kg.
- **75 mm thick.** That one's fixed.

- Face around 535 × 300 mm. We're flexible on the exact face. Pick what fits your plates and still gives 12 L. Whatever we choose becomes our standard block for every freezer after this, so let's agree it before plates get ordered.
- We talked about two dividers per station, three blocks across. Check that works with your plate height.
- 90 minutes is our guess for 75 mm. Please check it and give us the real number for whole squid and for salmon paste.
- **Drainage, this one matters.** Whole squid dumps a lot of water and ink when you load it. The bottom of the station has to let that liquid out, so the customer gets a block of squid and not a block of ice. But the paste can't leak out. We need a bottom that handles both. Happy to send photos or samples.

THE FREEZERS

- Four identical vertical plate freezers, indoors, top-loaded. Hydraulic open/close and lift, hot-gas release, blocks come out one side onto a table or conveyor.
- Loading is manual or semi-automatic to start. We're thinking a conveyor with gates for whole squid and a hose/pump for the paste; your ideas are welcome. But build them so your automatic loading and unloading can bolt on later: clearance on top, mounting points, spare control I/O.
- Blocks go into a bag, then a box. You mentioned a partner's bagger and box sealer; quote it as an option.
- Materials: aluminum plates. Stainless or aluminum anywhere product or wash water touches, nothing galvanized above the plates. For the base frame, give us stainless and hot-dip galvanized with the price difference. It's a salt-water, chlorine-washdown plant.
- Hydraulic unit: 460 V / 60 Hz motor, placed out of the cold and wet area, happy with food-grade (H1) oil, rated for −35 °C at the cylinders. Tell us if it's one unit per freezer or one shared.
- Stainless, ammonia-rated flex hoses that can be swapped from outside. No copper or brass anywhere near the ammonia.
- E-stops, guarding at the open top, a lockable disconnect. APCCO will check it against US rules.
- English manuals, drawings, electrical schematics, a two-year spares list, and STEP models so we can lay out the room.

REFRIGERATION

- **Ammonia (R717).** California basically rules out Freon at this size.
- Wolfking supplies the whole package: compressors, condenser, vessels, pumps, valves, controls. APCCO runs the piping and wiring on site.
- We like the **containerized engine room** idea: outside, next to the building, freezers inside, insulated pipes between. Saves space and keeps the noise outside. Joe and APCCO are meeting at the plant in early November to pick the spot, so send your standard container dimensions before then. Joe also asked about a small self-contained unit on each freezer; tell us if that's realistic with ammonia and what it would cost.

- **Keep the ammonia charge as low as you can.** In California, 500 lb (227 kg) on site is where the extra paperwork (CalARP) starts. If we can't get under that at this size, fine, just tell us the number.
- APCCO's rough early estimate, before we went to 140 blocks: about 45 TR at −30 °F (−34 °C) suction, pumped recirculation. Please do your own load calc for the final batch size.
- Tell us: system type and recirculation ratio; suction and condensing temperatures; condenser type (evaporative or air-cooled; water and noise both matter here); receiver or not; defrost method (hot gas?) and what the freezers need for it; which valves come mounted, which come loose, which you expect APCCO to supply, with brands.
- Vessels need the ASME U-stamp, piping to ASME B31.5, and the plates have to be rated for discharge pressure because hot gas hits them. APCCO will give you the numbers. California wants weld X-rays on some vessels; APCCO will say which.

POWER

- 480 V, 3-phase, 60 Hz. Motors nameplated 460 V / 60 Hz. There's no 380 V and no 220 V single-phase in the US.
- Controls on 120 VAC and 24 VDC.
- Panels built to UL 508A (UL or ETL listed), NEMA 4X stainless where there's washdown, name-brand UL components (Schneider, ABB, Siemens, Eaton, Allen-Bradley).
- Motors NEMA Premium / IE3. The US requires it for imports, even inside a machine. Size pumps and fans for 60 Hz speed, not relabelled 50 Hz units.
- **Send a load list early:** kW, amps and starting current per freezer and per skid. We need it to apply to the power company, and that can take longer than building the machines.
- PLC and screens in English. Remote access for your support is fine. Siemens or Allen-Bradley is easiest to service here.

THE SITE

- Watsonville, California, on Monterey Bay, about 150 km south of San Francisco. Mild coastal weather; APCCO will send design temperatures for the condenser.
- Earthquake country (Loma Prieta 1989 was right here). Everything gets anchored. Send weights, centres of gravity and anchor-bolt locations; APCCO's engineer does the calculations. Footprints and floor loads too.
- Ship to Oakland or Long Beach, then truck to Watsonville. Keep the engine room package within container or road-legal size.

CALIFORNIA THINGS TO KNOW WHILE YOU DESIGN

The short version. APCCO is the authority on all of this and will give the exact values.

- Permits go through the City of Watsonville on drawings stamped by APCCO's California engineer. Permits can take longer than manufacturing, so drawings come first.

- Ammonia design to IIAR 2 (and IIAR 4/5/6 for install and start-up) plus the California Mechanical Code: design pressures, relief valves, valve labelling, emergency shutdown, gas detection and ventilation for indoor machinery rooms. An outdoor package avoids most of the indoor part.
- 500 lb of ammonia triggers CalARP; 10,000 lb triggers federal PSM/RMP.
- ASME U-stamp vessels, B31.5 piping, Cal/OSHA vessel permits with weld radiography where required. Material certs and welder qualifications on file.
- Every electrical item needs a US listing (UL, ETL or CSA). Unlisted gear means a field evaluation, which is slow and expensive.
- Seismic anchorage per the California Building Code.
- It's a food plant under FDA rules: cleanable, corrosion-resistant surfaces, H1 lubricants, no flaking coatings over product.
- Cal/OSHA guarding, E-stops and lockout. Noise limits at the property line for outdoor gear.

TIMING

- **Now:** these notes; Joe and APCCO meet at the plant on the layout; Wolfking starts.
- **As soon as you can:** preliminary drawings, P&ID, load list, ammonia charge estimate and a proposal. The permit work starts from the preliminary design.
- **Build:** about 100 days after design approval, per your estimate at the show.
- **Running by May 2027,** June at the latest. Joe told the pet-food customer six months.
- **2027–28:** automatic loading and unloading, bagging, boxing, check-weighing.

WHAT WOULD HELP US MOST RIGHT NOW

1. A preliminary layout of one freezer: size, plate count, your block-face proposal, loading height, discharge side, weight.
2. A freeze-time check for 75 mm whole squid and salmon paste.
3. The refrigeration concept: system type, P&ID, equipment list, ammonia charge, container dimensions.
4. Electrical load list and one-line.
5. Vessel list with ASME status, and which certificates you can supply.
6. A proposal: what's Wolfking's and what's APCCO's, price, delivery, terms, spares, technician terms.
7. STEP models when you have them.
8. A couple of US references we can call (Oregon, LA).
9. A ballpark for the phase-2 automation and the bagger/boxer, so we design the interfaces now.

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中文

谁负责什么

- **Del Mar（Joe）**——客户。定冻块尺寸，批准平面布置和最终设计。
- **狼王**——设计并制造 4 台立式平板冻结机（含液压和控制），以及配套的氨制冷系统。提供图纸和办理许可证所需的资料，并派几位技术人员协助 APCCO 的施工队安装和开机。
- **APCCO**——我们在加州的本地制冷承包商。按加州法规审核设计，盖章并申报许可证，安装全部设备（管道、保温、电气），之后负责维护。他们不造冻结机，职责就是把设备装好、合规。他们会直接和狼王的工程师沟通，会问很多问题，这是办许可证的正常流程。
- 所有沟通都放在同一个邮件线程里，三方都抄送。

> **Sean，关于你说的波兰那件事**（货到港口才被要求补文件）：在开工制造之前，APCCO 会先给狼王一份清单，列出许可证需要的全部证书和文件，狼王告诉我们哪些能提供。这样到港口不会有意外。

我们要做什么

- **整条加州鱿鱼**，10 kg 冻块。这是面向中国市场的试水：如果中国买家接受平板冻的鱿鱼，我们会把更多产量转过来。蒙特雷湾鱿鱼季 5 月开始，所以我们希望 2027 年 5 月能开机。
- **三文鱼骨架绞成的鱼糜**（切片后剩下的鱼骨架，绞成糊状），供宠物食品客户，全年生产。这个合同就是用来付设备钱的。

45 t	2 × 10 h	4	140
每天（50 美吨）	两班 + 4 小时清洗	台立式平板冻结机	每台每批 10 kg 冻块数（期望值）

我们的算法：冻结 90 分钟 + 卸料装料 30 分钟 = 每 2 小时一批，每班 4 批，每天 8 批。45 t ÷ 8 ÷ 4 台 ≈ 每台每批 1.4 t，也就是 140 块。如果设计出来略少一点也没关系。

为什么做这个：现在我们是鱿鱼装纸箱、上架、进鼓风冻结间，人工太多，而这里人工是每小时 30 美元。这 4 台是试验。做得通，我们就按同样的设计继续加，之后再把装料、卸料、装袋、装箱自动化。

冻块

- 10 kg，约 12 升（12,000,000 mm^3）。鱿鱼比水轻 20% 左右，所以 12 L ≈ 10 kg。
- **厚度 75 mm。**这个是定的。
- 平面尺寸大约 535 × 300 mm。具体尺寸可以灵活，你们按自己的冻结板来定，保证 12 L 就行。但选定之后就是我们今后所有冻结机的标准冻块，所以订板之前先定下来。
- 我们谈的是每格两块隔板、并排三块冻块。请核对一下和你们的板高是否匹配。
- 75 mm 冻 90 分钟是我们的估计。请你们核算一下，给出整鱿鱼和三文鱼糜各自的真实时间。
- **排水，这点很重要。**整鱿鱼装进去会出很多水和墨汁。格子底部得能把这些水放出去，客户拿到的才是一块鱿鱼而不是一块冰。但鱼糜又不能漏出去。需要一个两样都兼顾的底部设计。需要的话我们可以寄照片或样品。

冻结机

- 4 台一样的立式平板冻结机，室内安装，顶部装料。液压开合板和顶升，热氨脱模，冻块从一侧出到工作台或输送机上。
- 装料一开始是人工或半自动。我们在想整鱿鱼用带闸门的输送机、鱼糜用软管加泵，欢迎你们出主意。但请把今后加装你们自动装料/卸料的条件预留好：顶部空间、安装点、备用控制 I/O。
- 冻块先装袋再装箱。你们提到合作伙伴的装袋机和封箱机，请作为选项一起报价。
- 材料：铝合金冻结板。凡是产品和冲洗水碰得到的地方用不锈钢或铝，冻结板以上不要有镀锌件。底座机架请分别报不锈钢和热镀锌，并给出差价。我们厂里是海水加含氯冲洗的环境。
- 液压站：电机 460 V / 60 Hz，放在低温潮湿区域之外，能用食品级（H1）液压油，油缸处耐 −35 °C。请告诉我们是每台一套还是几台共用。
- 不锈钢氨用软管，能从外部更换。氨系统附近任何地方都不要用铜或黄铜。
- 急停、顶部开口处的防护、可上锁的电源开关。APCCO 会按美国的规定检查。
- 英文的手册、图纸、电气原理图、两年备件清单，还有 STEP 三维模型，方便我们排布车间。

制冷

- **氨（R717）。**这个规模在加州基本不能用氟利昂。
- 狼王供整套：压缩机、冷凝器、容器、泵、阀门、控制。现场管道和接线由 APCCO 做。

- 我们喜欢**集装箱式机房**的思路：放在室外紧挨厂房，冻结机在室内，中间走保温管。省地方，噪音也在外面。Joe 和 APCCO 11 月上旬在厂里碰头定位置，请在那之前把你们标准集装箱的尺寸发过来。Joe 还问过每台冻结机配一个小型独立机组的方案，请告诉我们用氨做是否现实、大概多少钱。
- **氨充注量尽量做小。**在加州，现场 500 磅（227 kg）是额外手续（CalARP）的起点。如果这个规模做不到那么小也没关系，告诉我们数字就行。
- APCCO 早先的粗估（那时还没提到 140 块）：4 台约 45 冷吨，蒸发温度 −30 °F（−34 °C），泵循环。请你们按最终的每批冻块数自己算一遍负荷。
- 请告诉我们：系统形式和循环倍率；蒸发和冷凝温度；冷凝器类型（蒸发式还是风冷，这里用水和噪音都有讲究）；带不带贮液器；融霜方式（热氨？）以及冻结机对此的要求；哪些阀门随机装好、哪些散件、哪些你们希望由 APCCO 供，附品牌。
- 容器要有 ASME U 钢印，管道按 ASME B31.5，冻结板要按排气压力设计，因为热氨会打到板上。具体数值 APCCO 会给。加州对部分容器要求焊缝拍片，哪些需要 APCCO 会说。

电

- 480 V，三相，60 Hz。电机铭牌 460 V / 60 Hz。美国没有 380 V，也没有 220 V 单相。
- 控制用 120 VAC 和 24 VDC。
- 控制柜按 UL 508A 做（UL 或 ETL 认证），有冲洗的地方用 NEMA 4X 不锈钢柜，元件用带 UL 的大品牌（施耐德、ABB、西门子、伊顿、Allen-Bradley）。
- 电机 NEMA Premium / IE3。美国对进口电机有这个要求，装在整机里也一样。泵和风机按 60 Hz 转速选，不要拿 50 Hz 的换铭牌。
- **负荷清单请早点给：**每台冻结机、每个撬块的 kW、电流和启动电流。我们要拿它去电力公司申请用电，这个可能比造机器还慢。
- PLC 和触摸屏用英文。你们远程接入做支持没问题。西门子或 Allen-Bradley 在这边最好修。

现场

- 加州沃森维尔（Watsonville），在蒙特雷湾，旧金山以南约 150 km。沿海气候温和，冷凝器的设计温度 APCCO 会发。
- 地震区（1989 年洛马普列塔地震就在这里）。所有设备都要锚固。请提供重量、重心和地脚螺栓位置，计算由 APCCO 的工程师做。占地尺寸和地面荷载也一并给。
- 海运到奥克兰或长滩，再卡车到沃森维尔。机房包装请控制在集装箱或公路合法尺寸内。

设计时要知道的几条加州规定

简要版。这些都以 APCCO 为准，具体数值他们会给。

- 许可证由沃森维尔市政府签发，图纸要由 APCCO 的加州注册工程师盖章。许可证可能比造机器还慢，所以图纸先行。
- 氨系统按 IIAR 2 设计（安装和开机按 IIAR 4/5/6），再加《加州机械规范》：设计压力、安全阀、阀门标识、紧急停机、室内机房的气体检测和通风。室外机房可以省掉大部分室内的要求。
- 氨 500 磅触发 CalARP；10,000 磅触发联邦 PSM/RMP。
- 容器 ASME U 钢印，管道 B31.5，Cal/OSHA 容器许可证，需要时焊缝拍片。材质证明和焊工资质要留档。
- 所有电气件都要有美国认证（UL、ETL 或 CSA）。没认证的要现场评估，又慢又贵。
- 按《加州建筑规范》做抗震锚固。
- 这是 FDA 管的食品厂：表面易清洗、耐腐蚀，润滑油用 H1 级，产品上方不要有会掉皮的涂层。
- Cal/OSHA 的防护、急停和上锁要求。室外设备在地界处有噪音限值。

时间

- **现在：**这份备忘；Joe 和 APCCO 在厂里碰头定布置；狼王开始。
- **尽快：**初步图纸、P&ID、负荷清单、氨充注量估算和报价。许可证工作从初步设计就开始。
- **制造：**设计批准后约 100 天，按你们在展会上的估计。
- **2027 年 5 月开机，**最晚 6 月。Joe 跟宠物食品客户说的是六个月。
- **2027–28 年：**自动装料卸料、装袋、装箱、检重。

现在最需要你们提供的

1. 一台冻结机的初步布置图：尺寸、板数、你们建议的冻块平面尺寸、装料高度、出料侧、重量。
2. 75 mm 整鱿鱼和三文鱼糜的冻结时间核算。
3. 制冷方案：系统形式、P&ID、设备清单、氨充注量、集装箱尺寸。
4. 电气负荷清单和单线图。
5. 容器清单（含 ASME 情况），以及你们能提供哪些证书。

6. 报价：狼王和 APCCO 各管什么、价格、交期、条款、备件、技术人员条件。
7. STEP 三维模型，有了就发。
8. 一两个我们能联系的美国客户（俄勒冈、洛杉矶）。
9. 第二阶段自动化和装袋/装箱机的大概预算，好让我们现在就把接口设计进去。

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